Nolifa
Project breakdown

What we are building, who it is for, what it stands for, and how it answers the problem: how might communities use data and digital services to prepare for disruptions, reduce household costs and improve access to water, energy, transport and food?

One product. Three doors: Grow, Drop, Delivery.

1. Who Nolifa is

When disease, drought or an outage hits, a farm and its neighbours stay fed and watered.

Nolifa is one product with three doors. A farmer opens Grow, a household opens Drop, a driver opens Delivery. They share one engine, so a farmer, a customer and a driver see the same order, the same water warning and the same rules.

The promise: Protect, Prepare, Provide

Protect	Prepare	Provide
Keep farms healthy. Scan a leaf or check a sick animal. See outbreaks on a map before they reach your fence.	See trouble coming. Water forecasts, weather and supply warnings, and a way to ask for help together.	Keep food and water moving. Local farms sell to the people nearby, and the order is brought to the door.

Who it is for

- **The farmer first.** A smallholder, often in a rural area, often with poor signal and little money. Grow is the heart of Nolifa.
- **The household.** A family that wants steady food and water, and a warning before taps run dry.
- **The neighbour with a bike or a bakkie.** A driver who earns by bringing food to the people around them.
- **The community.** The shared water voice, the outbreak map and the community alerts belong to everyone in an area.

What we stand for

Value	What it means in the product
Free to protect	Scanning, symptom checks, the water radar, alerts and the community are free for everyone. Only selling has a plan
Proof, not talk	A community post needs a photo and an AI check. A delivery needs a PIN. A farm needs inspection
Honest about limits	The app says what is an estimate, what is simulated and what needs a connection or another person
Local first	KwaZulu-Natal conditions, local vets, English and isiZulu, built for patchy signal and low-end phones
Fair to rural farms	Inspection is tiered, so a remote farm can start earning without waiting for a surveyor

How Nolifa sounds

- Plain and direct. Short sentences. No jargon. A farmer reading on a cracked screen should understand it the first time.
- Warm but serious. It warns clearly and never panics. Urgent means urgent, Watch means watch.
- Respectful of effort. It speaks to a farmer as someone who knows their land and is getting a second pair of eyes.
- Never over-claims. The AI is a guide, not a diagnosis, and the app says so.

How Nolifa looks

A dark, calm, glass-like interface with deep forest green, a bright sprout green for good news and action, amber for watch and warmth, and red only for urgent. Emoji and icons carry meaning, so the app works for people who read little. A light appearance is available. Farm cards use real photographs and an emblem logo for each farm.

Night #0B1F17 Forest #1B4332 Sprout #52D1A4 Amber #E8A33D Urgent #FF6B6B



Farm emblems generated for the demo farms. A real farm gets its own.

2. How Nolifa answers the problem

Problem statement: how might communities use data and digital services to prepare for disruptions, reduce household costs and improve access to water, energy, transport and food?

Part of the problem	Our answer	How strong
Prepare for disruptions	Water forecast, outbreak map, low-water pop-up, household readiness score, outage bundles, notifications	Good
Water	Community radar, vote, provider desk, farm suppliers, bottle planner	Strong
Food	Local farm shop, escrow, delivery PIN, rescue deals for surplus produce	Strong
Weather and crop threats	Real 7-day forecast (Open-Meteo) turns into frost, heat, heavy rain, dry spell and wind alerts that name the farmer's own plots. The irrigation AI plans each plot's watering from the weather, soil and crop	Built, field test needed
Water data	Real dam levels from the Department of Water and Sanitation (KZN weekly report) with a rain-corrected forecast. Tap-level outages are not available anywhere yet; road map in docs/WATER_ROADMAP.md	Real for dams, roadmap for taps
Road disruption	Drivers warn each other about potholes, floods, crashes and roadblocks on the job map, like Waze	Built, shared only with the backend
Proof it works	An Impact page that measures accuracy, forecast checks, harvest dates and water saved, with sample sizes and ranges	Built, needs real use
Household costs	My savings page: rescue deals, group buys, fees paid and net saving, plus deals to take now. No price comparison yet	Good
Energy	Outage bundles (gas, candles, power bank). No outage schedule or sharing	Roadmap
Transport	Neighbour delivery moves goods. No help for people travelling	Roadmap

We would rather show water and food in depth, honestly, than claim all four resources. Energy and transport are roadmap, built on the same structure.

3. The three doors

	Grow	Drop	Delivery
For	Farmers	Households	Drivers
Opens on	Scan a crop, check an animal	Farms near you and a water warning	Jobs nearby
Tabs	Home, Orders, Alerts, Settings	Shop, Water, Alerts, menu	Jobs, Earnings, Account
Unique	Crop health AI, hexagon outbreak map, proof-backed community, water supplier	Farm cards with photos, cart, escrow, readiness score	Route map, navigation, proof photos, PIN delivery

4. One order, end to end

Step	What happens
1	A farm registers, is verified, pays a plan, shows its land with photos and lists stock

Step	What happens
2	A household browses farm cards, fills a one-farm basket and pays into escrow (simulated in the prototype)
3	The farmer is notified, packs the order and chooses to deliver or request a driver
4	The driver navigates to the farm, takes a pickup photo, follows the route to the customer and takes a delivery photo
5	The customer gives the 4-digit PIN. Delivery completes
6	The customer confirms, or 2 hours pass, and escrow pays the farm and driver. A problem freezes the money and opens a dispute

5. What each door does

Nolifa Grow (farmer)

- **Home:** recent scans, Scan a crop, Crop health AI (plots, growth stage, days to harvest), Irrigation AI, Check a sick animal, Your herd, Water radar, Water provider.
- **Livestock:** cattle, goats, sheep and horses, each with its own classes, breeds and diseases. Each animal has a health record, a location pin and a state on a herd map that flags groups of sick animals.
- **My farm:** size in hectares, soil, water sources (tanks with levels), irrigation systems and which plot uses which.
- **Irrigation AI:** a daily water balance per plot (FAO-56). It says water or wait, how many litres and minutes, the week ahead, tank days left, and why. A soil check (dry, fine, wet) sharpens it. With a smart valve link it can open the valve in the morning while the app is open.
- **Impact evidence:** scan verdicts with a 95% range, forecast against observed weather, harvest against prediction, and water saved by rain, exportable as CSV.
- **Orders:** earnings, orders to send, notifications, stock, selling status, delivery choice.
- **Alerts:** red alarm for the farm's own warnings, hexagon outbreak map, Report an outbreak, water forecast, Water radar, Community alerts (neighbour reports, area lists, proof-backed chat).
- **Settings:** details, farm info, language, appearance, business setup, wallet, trust and safety.

Nolifa Drop (household)

- Shop laid out like a food app: address, cart, search, carousel, farm cards with photos and offerings. One farm per basket.
- Water tab with forecast, vote and the source of the numbers. Alerts tab. Menu with orders, wallet, settings, trust, help and an offline guide.
- Welcome back notification and a low-water pop-up with a Vote button when the app opens.

Nolifa Delivery (driver)

- Verified drivers see jobs, accept one, and get a driving route (OSRM) from their position to the farm and the customer, checked on the simulator.
- **Road alerts:** one tap reports a pothole, flood, crash, roadworks, roadblock, lights out, blocked road or animals. Warnings ahead show with distance, and near one the app asks still there or gone. Warnings fade unless confirmed. Only verified drivers report.
- Navigate opens Apple or Google Maps. Pickup and delivery open the camera for proof. The customer's PIN completes the delivery.
- Drivers keep 80% of the delivery fee. No welcome notification, by design.

6. Rules that build trust

Rule	Detail
Buyer fee	3% of items, min R2, max R15
Delivery	R25 base on Nolifa Delivery. A farm delivering itself sets its own fee. Driver keeps 80%
Farm plan (to sell)	Starter R49 (8% commission), Grow R99 (5%), Co-op R199 (3%). Prices are hypotheses to test
Selling	Register, verify, plan, show land, then sell fresh produce with limits (R2 000, 45 days) while an inspection is arranged. Eggs, milk and processed goods wait for it
Inspection routes	Surveyor (about 14 days), extension officer (about 7), co-op day (about 21)
Escrow and PIN	Money is held until the customer gives the PIN or confirms. Disputes freeze it

7. The AI, and what is honest about it

Feature	What it is	Limit
Leaf scanner	Trained model on the phone, with an online service when there is signal	A guide, not a lab test. Accuracy not yet measured on our own photos
Animal checker	Symptom matcher for cattle, goats, sheep and horses: about 40 signs, 40 conditions, controlled diseases flagged first, no doses	Not a vet. Not validated by a vet yet
Irrigation AI	FAO-56 soil water balance with crop factors, real evaporation and rain. Rule-based, shows how sure it is	Cannot sense soil. Assumes half used until told. Autopilot needs the app open and a smart valve
Crop health AI	Crop calendar adjusted by scans and watering, with a range and confidence	An estimate, not a neural network
Water forecast	Smoothing on the weekly dam readings, corrected with forecast rain, confidence capped	Two weekly points only. Cannot predict a burst main

8. What is real, what is simulated, what we still owe

Real in the prototype	Simulated or not built
Apps and on-phone storage, fee and escrow rules, PIN, disputes logic, inspection rules, crop advisor, hexagon map, community proof rules, notifications, security rules and tests (about 470 across the projects: core 196, Grow 211, Drop 36, Delivery 28), real DWS dam levels, road warnings, irrigation maths checked by hand	Payments and escrow, identity and bank checks, inspection results, water provider replies, tap-level water outages, neighbour and driver activity, sharing of road warnings and records between phones, push notifications, running irrigation with the phone locked. Rules tests for roads are written but unrun (no Java)

Evidence still to collect

- Leaf scanner accuracy on 50 to 100 labelled photos from KZN.
- Ten farmer interviews on usefulness and willingness to pay.
- A small co-op trial with real orders and a real water-vote loop.
- The kit for this is in docs/EVIDENCE_KIT.md, with a script that scores the scanner on labelled photos.

- A sourced figure for the number of smallholder farmers. Do not claim free or 2.5 million without a source.

9. Gaps and roadmap

Item	Why	Size
Vet review of the four-species disease list	Safety of the advice	Small
Soil sensors or a farm hub for background irrigation	Removes the app-open limit	Medium
Municipal water feeds and household outage reports	Tap-level water	Medium
Field test of the weather alerts with real farmers	Shows they are useful and not noisy	Small
Price comparison across farms	Finishes the household cost story	Small
Outage schedule and power sharing	Energy	Medium, roadmap
Shared delivery runs and lift-share	Transport	Medium, roadmap
Shared backend, payments, identity, push	Makes the simulated parts real	Large, backend team

10. Security

Risk	What Nolifa does	Status
Delivery PIN can be guessed	3 tries. The third wrong try locks the order, opens a dispute and freezes the money	Built
Escrow runs on the device	Designed: a licensed provider (FNB) holds the funds and a server decides release. Simulated now	Designed
Personal data on the phone	AES-256-GCM encryption, key in the Keychain or Keystore. Older data is migrated	Built
One person, many accounts	Only verified people vote on water or confirm an outbreak. The backend must allow one verified number per account	Built, backend needed

Security rules for the PIN lock and verified voting are written and tested in code but the Firestore emulator tests have not been run yet. See docs/SECURITY.md.

11. Decisions for the team

- **The one sentence.** Do we agree on: when disease, drought or an outage hits, a farm and its neighbours stay fed and watered?
- **Who is the main user?** The farmer, with the household and driver carrying the rest. Confirm.
- **One product or three apps in the pitch?** Recommended: one product, three doors.
- **Scope.** Water and food in depth. Weather and savings next. Energy and transport as roadmap.
- **The demo.** One water disruption from warning to delivered food, with labelled test data and the provider's side.
- **Owners.** Who owns farm, water, shop, pitch and evidence.

- **Water votes.** Households vote once their phone is verified, not only verified farmers. Confirm.